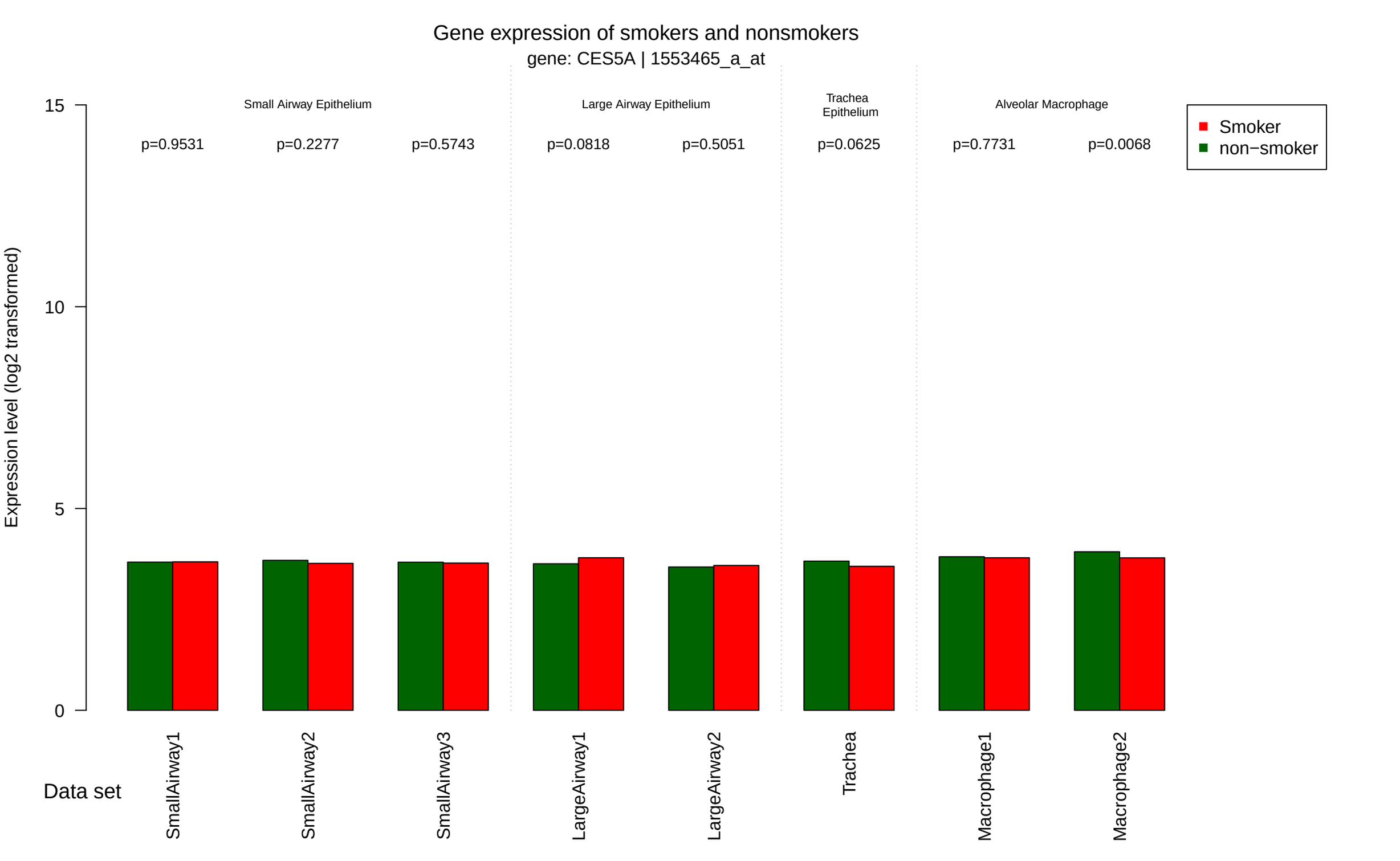
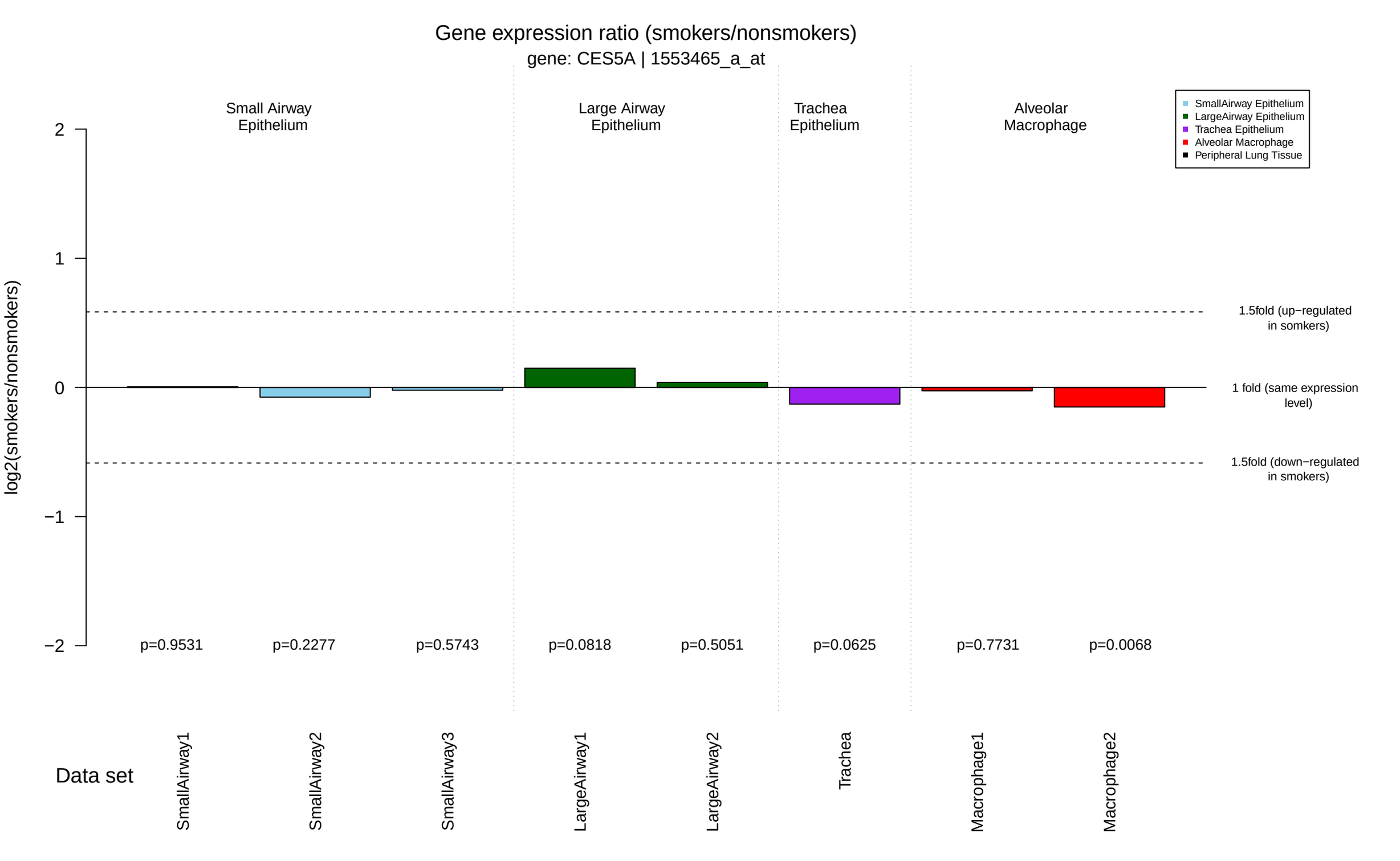
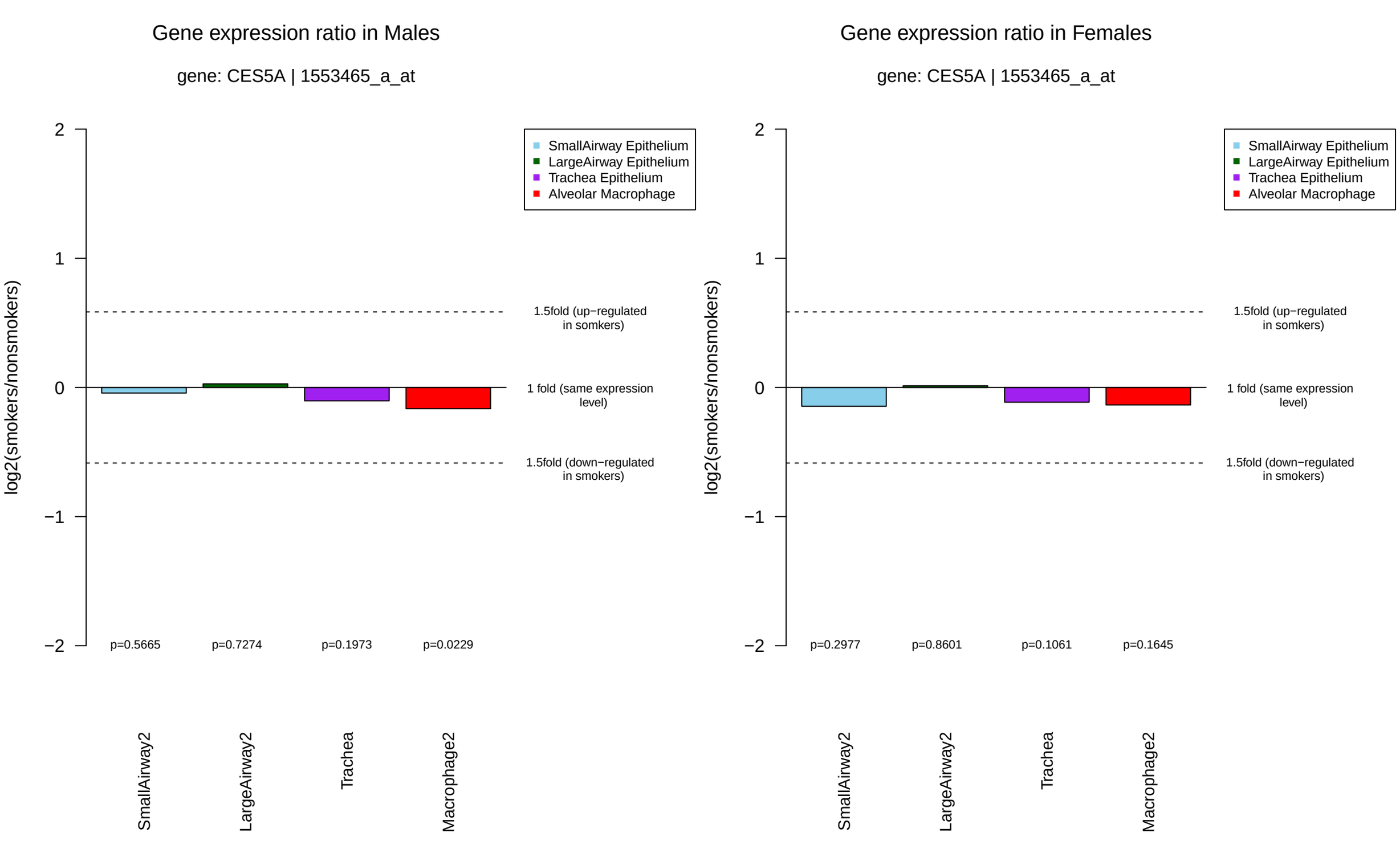
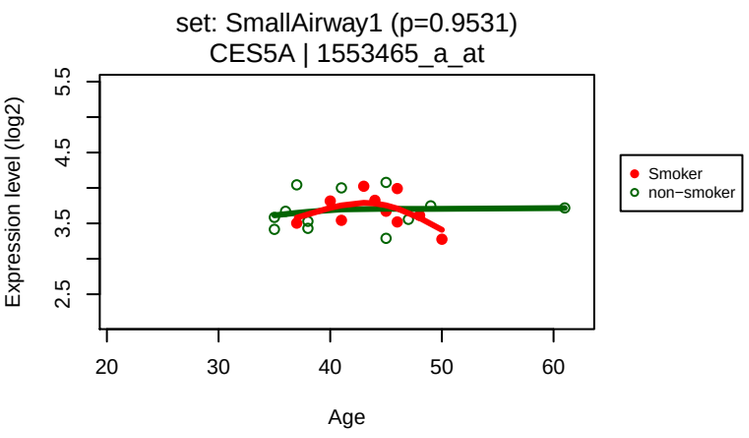
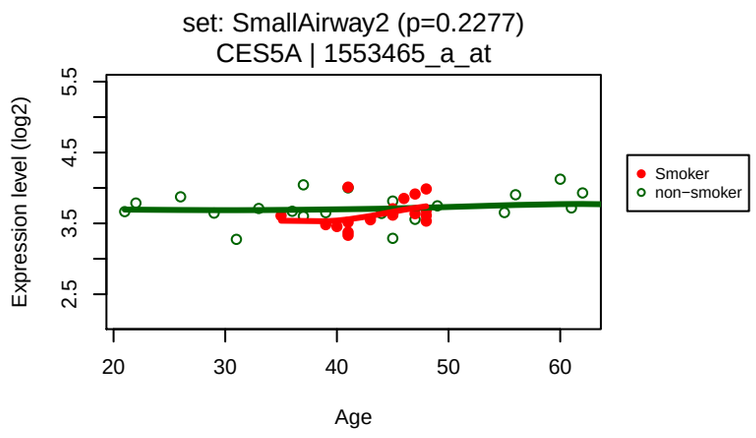
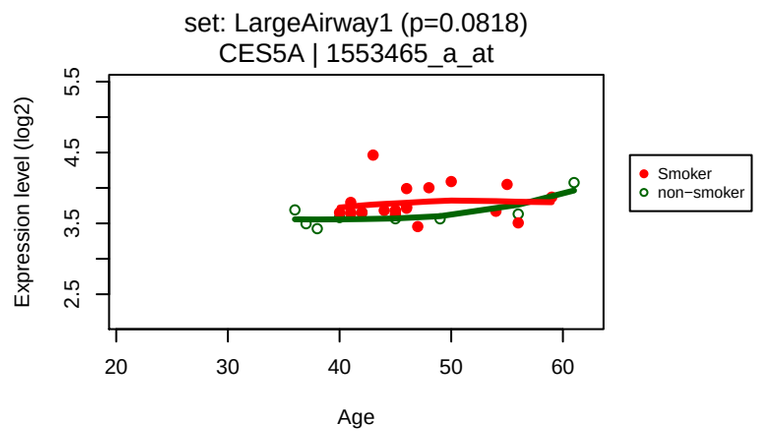

Gene expression of smokers and nonsmokers
gene: CES5A | 1553465_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9531
p=0.2277
p=0.5743
p=0.0818
p=0.5051
p=0.0625
p=0.7731
p=0.0068
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

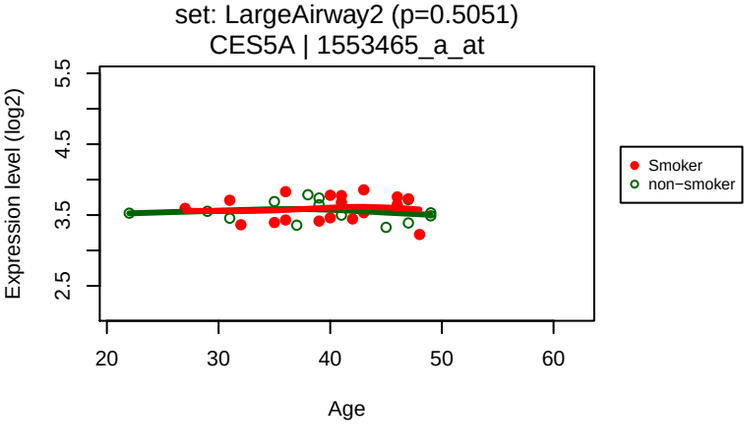

Gene expression ratio (smokers/nonsmokers)
gene: CES5A | 1553465_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9531
p=0.2277
p=0.5743
p=0.0818
p=0.5051
p=0.0625
p=0.7731
p=0.0068
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

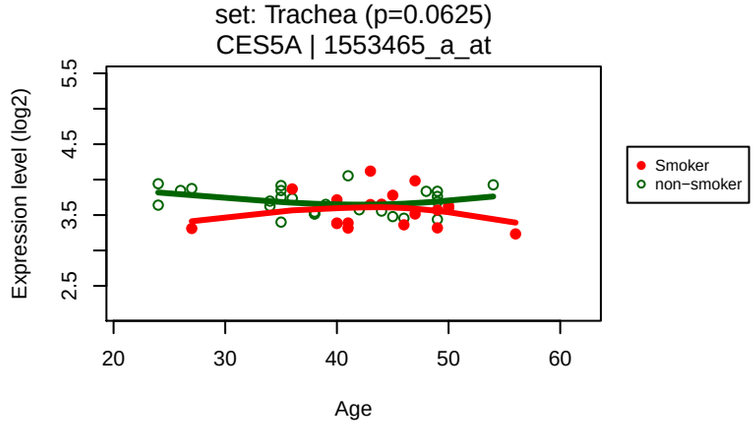

Gene expression ratio in Males
gene: CES5A | 1553465_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5665
p=0.7274
p=0.1973
p=0.0229
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CES5A | 1553465_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2977
p=0.8601
p=0.1061
p=0.1645
SmallAirway2
LargeAirway2
Trachea
Macrophage2

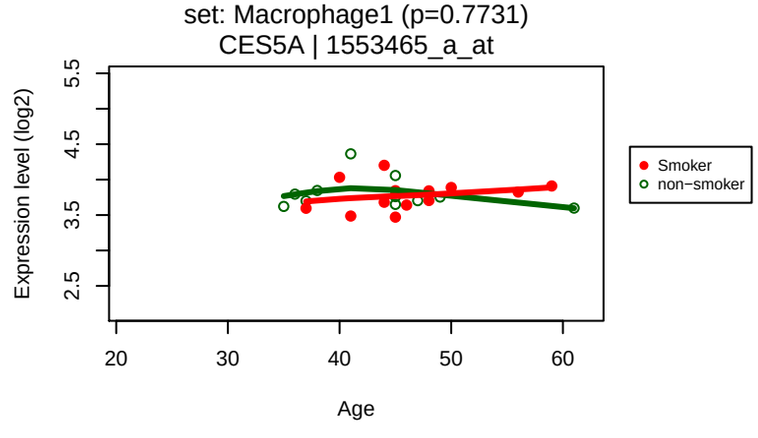

set: SmallAirway1 (p=0.9531)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker

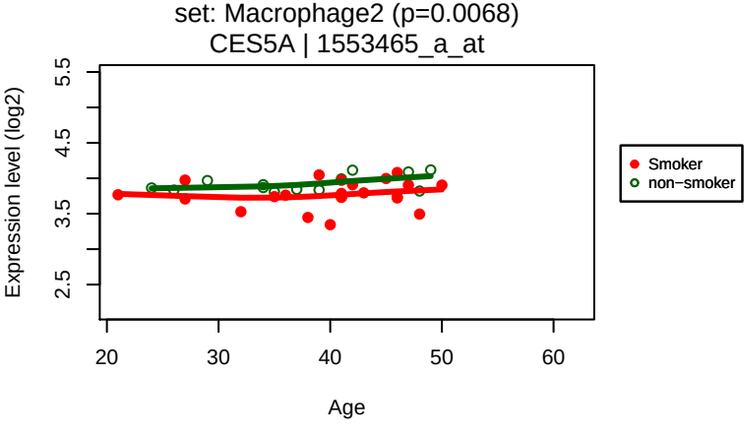

set: SmallAirway2 (p=0.2277)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker

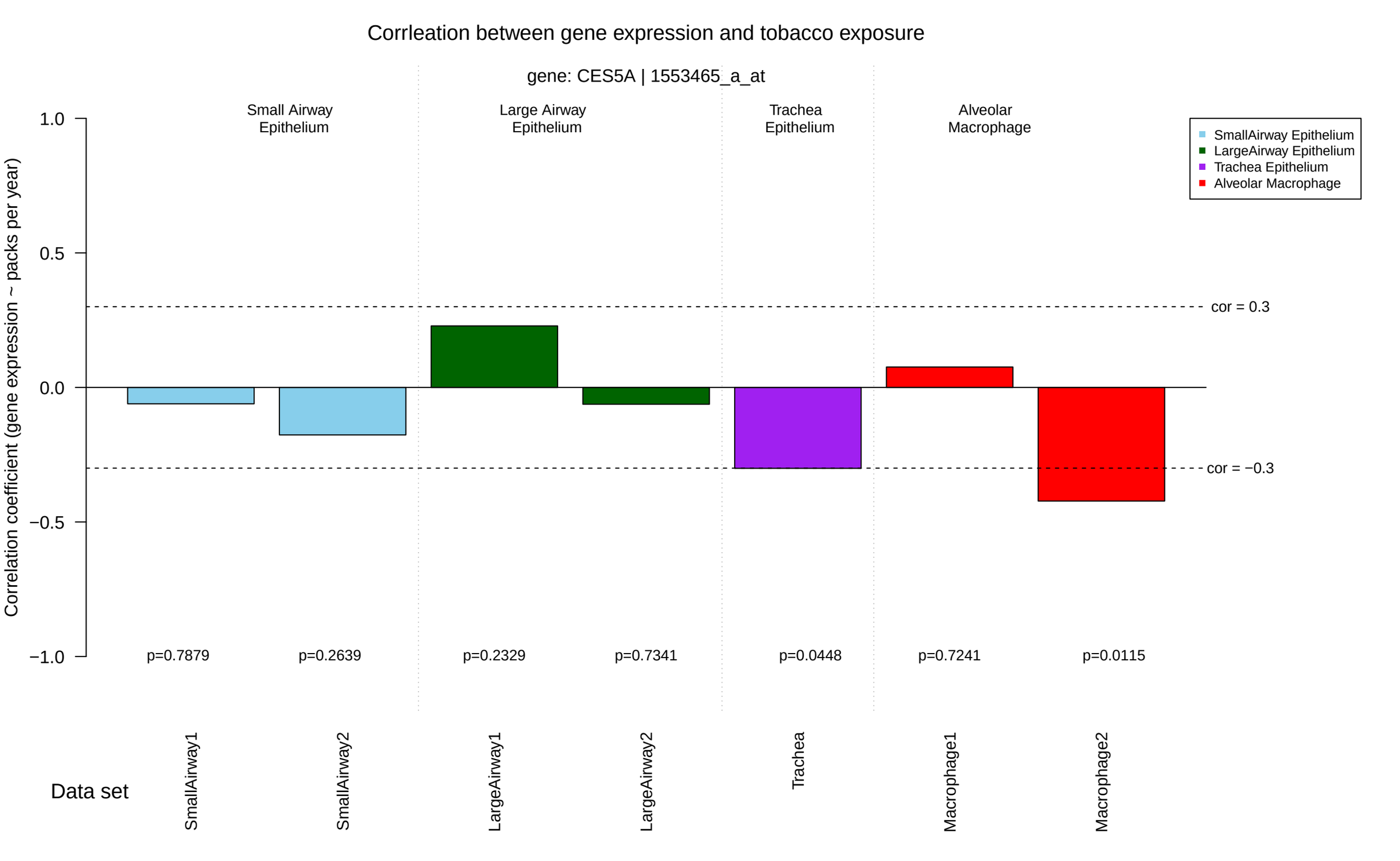

set: LargeAirway1 (p=0.0818)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5051)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0625)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7731)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0068)
CES5A | 1553465_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CES5A | 1553465_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7879
p=0.2639
p=0.2329
p=0.7341
p=0.0448
p=0.7241
p=0.0115
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2